SOCIOL 723: Midterm Exam, Answer Key

Thursday, October 15, 2026, 9:05–11:35AM

Instructions. You have 150 minutes. You can submit and leave early. The exam is open book and open note, and asks for no code. Circle or write your answers to Parts 1 and 2. Write Part 3 and the bonus in the space below each question; continue on the back of the page if you need more. Where a question asks for sentences, write them as you would in a paper. Show your arithmetic: a right number with no work earns partial credit only. No calculator is needed: the last page lists the exponentials and the harder products and quotients the exam calls for (and some it does not); write down the expression you need, then look up its value. The gray tag after each question names the slides it mainly draws on (Week 1 has two decks; "Tue" is Tuesday's).

Part	Format	Weeks	Points
1	Five yes-or-no questions	1–7	15
2	Ten single-choice questions	1–7	30
3	Three longer questions	1–2, 3–4, 5–7	55
Bonus	One derivation	5	+10

Part 1: Yes or No

Circle YES or NO. Three points each.

1. You regress log earnings on years of schooling and leave out a variable that raises earnings but is uncorrelated with schooling. This biases the OLS schooling slope. [Week 1 Tue, slide 33–35] YES NO

   Answer. NO. Bias $= \beta_2 \, \delta_1$, the effect of the omitted variable times its regression slope on schooling. With $\delta_1 = 0$ the bias is zero; the omitted variable only sits in the error and widens it.

2. A minimum-wage law is set at the state level, the kind of design Bertrand, Duflo, and Mullainathan (2004; Week 2, slide 42) studied, and your data are individuals in counties within states. Clustering the standard errors by county is appropriate. [Week 2, slide 40–41] YES NO

Answer. NO. Cluster at the level at which treatment was assigned, the state. Everyone in a state shares the same law and the same state-level shocks, so county clusters still treat correlated people as independent and understate the standard error.

3. In a logit of voting on education, you add age. The likelihood-ratio statistic for adding age is 3.0 (one extra parameter). AIC prefers the model with age. [Week 3, slide 69–70] YES NO

Answer. YES. AIC $= -2\ell + 2k$. Adding age lowers -2ℓ by $LR = 3.0$ and raises the penalty by 2: net -1.0, so the model with age has the lower AIC. (BIC, with $\log n$ per parameter, would say no for any $n > 20$.)

4. Test scores and athletic talent are independent among applicants, and a college admits everyone whose score plus talent clears a threshold. Among the *rejected* applicants, scores and talent can be associated. [Week 5, slide 35–36] [Lab 5, p. 8] YES NO

Answer. YES. Admission is a collider: scores $\rightarrow$ admitted $\leftarrow$ talent, and conditioning on *either* value of a collider opens the path. Among the rejected, score plus talent is below the threshold, so a rejected applicant with a high score must have had low talent: a negative association, just as among the admitted.

5. Like Cheng et al. (2021; Week 6, slide 32), you build a propensity score for completing college from pre-college covariates. Yours predicts who completes college and who does not almost perfectly. This is good news for matching. [Week 6, slide 30–31] YES NO

Answer. NO. A near-perfect propensity model means treated and untreated units sit at different values of $\hat{e}(x)$, so there is no overlap: no untreated unit looks like a treated one, and matching has nothing to compare. Good news would be a model that balances covariates while leaving substantial overlap.

Part 2: Single Choice

Circle one letter. Three points each.

1. You regress log earnings on schooling and age, and the schooling coefficient is 0.10, "the effect of schooling holding age constant." Which two-step procedure reproduces exactly 0.10? [Week 1 Tue, slide 26–29]

(A) Regress log earnings on schooling alone; age is already held constant by the data.

(B) Regress log earnings on age and keep the residuals; regress those residuals on schooling itself.

(C) Regress log earnings on age and keep the residuals; regress schooling on age and keep the residuals; regress the first residuals on the second.

(D) Regress schooling on age; the slope of that regression is 0.10.

Answer. (C). Frisch–Waugh–Lovell: strip age out of *both* log earnings and schooling, then regress residuals on residuals; the slope equals the multiple-regression coefficient exactly, in every sample. (B) is the common mistake: schooling itself still carries age, so the slope differs from 0.10 whenever schooling and age are correlated. (A) is the short regression, biased by omitted age. (D) is the auxiliary regression in the OVB formula, not the coefficient.

2. You replace the classical OLS standard errors with heteroskedasticity-robust (HC) ones. Which statement is correct? [Week 2, slide 19–21, 31]

(A) The robust standard error can be larger or smaller than the classical one; which happens depends on whether the errors are more spread out where X is far from its mean.

(B) The robust standard error is always at least as large as the classical one, because it allows for more.

(C) The coefficient changes slightly, because the robust formula downweights observations with large residuals.

(D) The robust standard error is always smaller than the classical one, because it uses the actual residuals instead of one pooled variance.

Answer. (A). No general direction: the sandwich weights squared errors by $(X_i - \bar{X})^2$, so the robust standard error exceeds the classical one when the errors are noisier at extreme X and falls short of it when they are noisier near the mean. On the slide the schooling SE went from 0.0051 (classical) to 0.0055 (HC3). The coefficient never changes.

3. In 40 independent Bernoulli trials there are 12 successes. Let θ be the probability of success on each trial. The maximum likelihood estimate $\hat{\theta}$ is [Week 3, slide 28]

(A) 0.70

(B) 0.40

(C) 12

(D) 0.30

Answer. (D). $\ell(\theta) = 12\log\theta + 28\log(1-\theta)$; the score $12/\theta - 28/(1-\theta) = 0$ gives $\hat{\theta} = 12/40 = 0.30$, the sample proportion.

4. Gelman, Fagan, and Kiss (2007, *JASA*) count the pedestrian stops the NYPD made of each ethnic group in each of 75 precincts over 15 months, about 125,000 stops in all. They fit an overdispersed Poisson regression of the number of stops with ethnic-group dummies, precinct effects, and the group's arrests in the previous year as the offset. (An overdispersed Poisson regression keeps the Poisson's mean, $\log\theta_i = X_i'\boldsymbol{\beta}$; the overdispersion term only widens the standard errors.) For violent-crime stops, the coefficient on Black (relative to white) is about 0.92, so $e^{0.92} = 2.5$. Which reading is right? [Week 4, slide 36–38]

   (A) Each Black resident was stopped 0.92 more times than each white resident.

   (B) Within a precinct, and relative to each group's arrests the year before, Black residents were stopped 2.5 times as often as whites.

   (C) The counts are overdispersed, so the 2.5 is biased upward and should be discounted.

   (D) Black residents were stopped 2.5 times as often as whites relative to their share of the precinct's population.

   Answer. (B). In a Poisson regression $e^{\hat{\beta}}$ multiplies the expected count; with the previous year's arrests as the offset, the count is a rate of stops per arrest, so the comparison is relative to arrests, not population (D), and the precinct effects make it a within-precinct comparison. (A) reads a log-scale coefficient as an additive count. (C) is wrong because overdispersion leaves the coefficients where the Poisson put them and only makes the plain Poisson standard errors too small; the authors' overdispersion term widens the standard errors, which is why they include it.

5. In a logit, the coefficient on years of experience is positive. One more year of experience changes the *probability* of a callback the most for applicants whose predicted probability is near [Week 4, slide 10, 14]

   (A) 0

   (B) 0.5

   (C) 1

   (D) It is the same change for every applicant.

   Answer. (B). The logit curve is steepest at $\hat{p} = 0.5$, where one unit of $X'\beta$ moves the probability by up to 0.25, and flat near 0 and 1. The coefficient is constant on the log-odds scale, not on the probability scale.

6. A school randomly assigns half of its ninth graders to an after-school tutoring program and, at the end of the year, compares the mean math score of tutored and untutored students. During the year, tutored students go on to help their untutored friends with homework. Which assumption fails, and with what consequence? [Week 5, slide 15]

   (A) Random assignment: students whose friends were tutored were more likely to be tutored themselves.

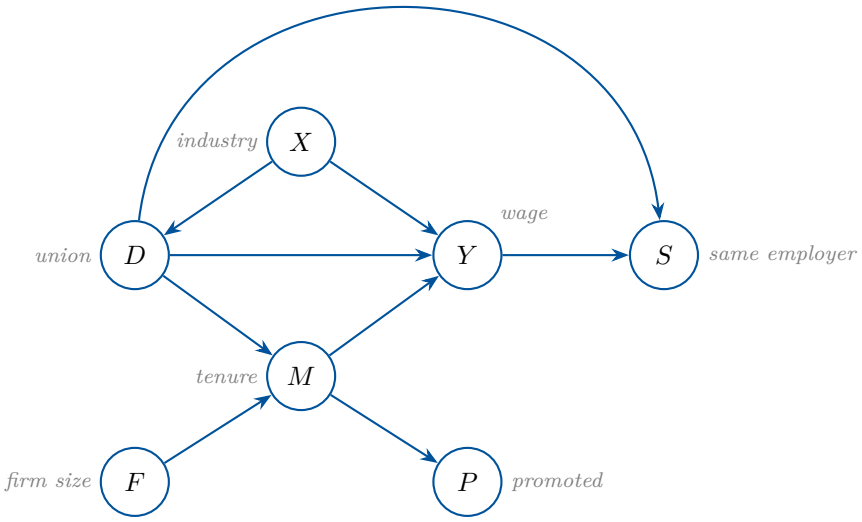
   (B) SUTVA: an untutored student's score depends on whether their friends were tutored, so the control group is partly treated and the comparison understates the program's effect on the tutored.

   (C) Positivity: students without tutored friends had no chance of being tutored.

   (D) None: assignment was random, so the difference in means estimates the ATE whatever the students do afterwards.

   Answer. (B). Interference: an untutored student's outcome depends on friends' treatment, so $Y_i(0)$ is not one well-defined number. Randomization is intact, which is why (A) and (C) are wrong, but it does not protect against spillovers: the untutored mean is pulled up by the help they receive, so tutored minus untutored understates the program's effect on those tutored.

7. In the DAG below, D is union membership, Y is the hourly wage five years later, X is industry, M is job tenure, F is firm size, P is whether the worker was promoted, and S is whether the worker is still with the same employer at follow-up. Which set satisfies the back-door criterion for the *total* effect of D on Y? [Week 5, slide 32, 41, 44–46]



(A) X and M.

(B) X and P.

(C) X only.

(D) X and S (that is, analyze only workers still with the same employer).

Answer. (C). X is the only confounder, on the back door $D \leftarrow X \rightarrow Y$, and blocks it by itself. F causes the mediator but lies on no path between D and Y; adding it would be harmless (slide 46, right panel) but is not needed. M is a mediator: holding it fixed removes the part of the effect that runs through tenure. P is a descendant of the mediator, a noisy copy of M, so it does the same damage diluted, and both violate condition 1 (no descendant of D). S is a collider ($D \rightarrow S \leftarrow Y$): keeping only workers still with the same employer opens a spurious path.

8. Exact matching on cells, as Mazrekaj, De Witte, and Cabus (2020; Week 6, slide 21) did for children of same-sex parents. Two cells of a covariate, with ignorability given the cell. Recall that the ATT weights each cell's treatment effect by the treated units' distribution across the cells. [Week 6, slide 11]

cell	treated (n)	untreated (n)	treated mean Y	untreated mean Y
young	20	40	12	10
old	30	10	20	14

The ATT is

(A) 3.6
(B) 4.0
(C) 4.4
(D) 6.0

Answer. (C). Within-cell differences $12 - 10 = 2$ (young) and $20 - 14 = 6$ (old). The ATT weights cells by their share of the *treated*: $(20 \times 2 + 30 \times 6)/50 = 220/50 = 4.4$. The ATE weights by everyone, $(60 \times 2 + 40 \times 6)/100 = 3.6$; 4.0 is the unweighted average; 6.0 is the naive comparison $16.8 - 10.8$.

9. You estimate the wage return to a college degree with the augmented IPW (doubly robust) estimator. It combines two fitted models: a *propensity model* $\hat{e}(x) = \widehat{\Pr}(D_i = 1 \mid X_i = x)$ for the probability of holding a degree given the covariates, and an *outcome model* $\hat{m}_d(x) = \widehat{E}[Y_i \mid D_i = d, X_i = x]$ for expected log wages given the covariates, one for graduates ($d = 1$) and one for non-graduates ($d = 0$). Which statement about the estimator is correct? [Week 6, slide 54, 60]

(A) It is consistent only if both the propensity model and the outcome model are correct.
(B) It never has extreme weights, because the outcome model stabilizes them.
(C) It removes bias from unobserved confounders because it uses two models.
(D) It is consistent if either the propensity model or the outcome model is correct, and it does nothing about unobserved confounders.

Answer. (D). AIPW is consistent if either the propensity score or the outcome model is right. It rests on conditional ignorability like everything else this week, and its weights are the same $1/\hat{e}$ weights, so extreme propensities still hurt.

10. Card (1995; Week 7, slide 29) instruments schooling D with growing up near a college, Z. Suppose the first stage is 0.2 years and that Z also raises log wages directly by 0.01 (a college town has a better labor market), a small violation of exclusion. In large samples the IV estimate of the return to schooling is off by about [Week 7, slide 21]

    (A) 0.01: the direct effect itself.

    (B) 0.002: the direct effect times the first stage.

    (C) 0.05: the direct effect divided by the first stage.

    (D) nothing: a direct effect this small is absorbed by the first stage.

    Answer. (C). $\operatorname{plim} \hat{\beta}_{IV} = \beta + \operatorname{Cov}(Z, \epsilon)/\operatorname{Cov}(Z, D)$. A direct effect of 0.01 enters $\operatorname{Cov}(Z, \epsilon)$ and is divided by the first stage: $0.01/0.2 = 0.05$. A weak first stage magnifies a small exclusion violation.

Part 3: Longer Questions

Points are shown with each question; 55 in all. Every number below comes from the course's GSS extract or from a paper discussed in Weeks 4, 6, and 7.

Question 1: Schooling, Vocabulary, and Earnings (20 points)

The GSS extract has 3,509 full-time workers aged 25–64 (GSS 2010–22). The outcome is log annual earnings; the vocabulary score is the number of ten words defined correctly, measured at the interview. Three OLS regressions, rounded to three decimals: columns (1) and (2) have log earnings as the outcome; column (3) is the auxiliary regression of the vocabulary score on schooling and female.

	(1) log earnings	(2) log earnings	(3) vocabulary score
years of schooling	0.124	0.108	0.271
female	−0.394	−0.392	−0.037
vocabulary score		0.062	
intercept	8.340	8.190	2.424
R^2		0.187	

Standard errors for the schooling coefficient in model (2), three ways: classical 0.0055; HC3 0.0056; clustered on the GSS primary sampling unit (839 clusters, median size four) 0.0056.

(a) Interpret the female coefficient in model (2), −0.392, in one sentence a reader of a sociology journal would understand. [Week 1 Tue, slide 23]

Answer. Among these workers, women earn about 0.39 log points less than men with the same schooling and vocabulary score, roughly 32% less ($e^{-0.392} - 1 = -0.32$). A description of the conditional mean, not yet a causal claim.

(b) Use the omitted-variable-bias formula to recover the schooling coefficient in model (1) from models (2) and (3). Show the arithmetic and sign the bias. [Week 1 Tue, slide 33–35]

Answer. Short = long + (effect of omitted) × (slope of omitted on included): $0.108 + 0.062 \times 0.271 = 0.108 + 0.017 = 0.125$, which is 0.124 up to rounding. Both pieces are positive (vocabulary raises earnings and goes with schooling), so the bias is positive: model (1) credits schooling with part of the vocabulary advantage.

(c) Vocabulary is measured at the interview, years after schooling ended, so schooling may have raised it. Draw a DAG with schooling, vocabulary, and earnings in which schooling raises vocabulary and vocabulary raises earnings. Under this DAG, is 0.108 the total effect of schooling on earnings? Explain in one or two sentences. [Week 1 Tue, slide 72] [Week 5, slide 46]

Answer. Schooling → vocabulary → earnings, with a direct arrow schooling → earnings. Vocabulary is a mediator. Holding it fixed removes the part of the schooling effect that works through vocabulary, so 0.108 is the direct effect, not the total effect. Under this DAG the total effect is model (1)'s 0.124 (given female), and adding the mediator is a bad control.

(d) The spread of log earnings is wider among the more schooled, so the errors are heteroskedastic. Does this bias 0.108? What does it change, and which of the printed standard errors answers it? [Week 2, slide 19–21, 30]

Answer. No. Unbiasedness needs only $E[\epsilon_i \mid X_i] = 0$ and says nothing about the variance of ϵ_i. Heteroskedasticity makes the classical formula for the standard error wrong, not the coefficient. Report the robust (HC3) standard error, 0.0056; here it barely differs from the classical 0.0055. The clustered standard error is also acceptable: the cluster-robust formula allows any heteroskedasticity as well.

Question 2: The Mark of a Criminal Record (Pager 2003; 15 points)

Pager (2003, *AJS*) sent matched pairs of testers to apply for 350 entry-level jobs in Milwaukee. The outcome is a callback (0/1). Testers were white or Black, and in each pair one presented a criminal record. Pager reports callback rates of 34% (white, no record), 17% (white, record), 14% (Black, no record), and 5% (Black, record). Her Appendix Table B1 fits a logit of callback on *record*, *Black*, and their product (intercept rounded; the table also carries two controls, so the model's predictions sit slightly off the raw rates):

	$\hat{\beta}$
intercept	−0.66
criminal record	−0.99
Black	−1.25
record × Black	−0.29

(a) Write the model twice: once in log-odds form and once in probability form. [Week 4, slide 9]

Answer. $\log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 \,\text{record}_i + \beta_2 \,\text{Black}_i + \beta_3 \,\text{record}_i \times \text{Black}_i = X_i'\boldsymbol{\beta}$, and $p_i = e^{X_i'\boldsymbol{\beta}}/(1 + e^{X_i'\boldsymbol{\beta}})$.

(b) Compute, by hand, the predicted callback probability for a white tester without a record and for a white tester with a record. Compare with Pager's 34% and 17%. [Week 4, slide 15–16]

Answer. No record: $X'\hat{\beta} = -0.66$, odds $e^{-0.66} = 0.52$, $\hat{p} = 0.52/1.52 = 0.34$. Record: $-0.66 - 0.99 = -1.65$, odds $e^{-1.65} = 0.19$, $\hat{p} = 0.19/1.19 = 0.16$. Pager's 34% and 17%; the one-point gap is her two controls.

(c) Read the record coefficient for white testers on all three scales: log-odds, odds, and probability. [Week 4, slide 14, 16]

Answer. Log-odds: a record lowers the log-odds of a callback by 0.99. Odds: it multiplies the odds by $e^{-0.99} = 0.37$, cutting them by 63%. Probability: for white testers the callback probability falls from 0.34 to 0.16, about 18 points, or roughly in half.

(d) Now compute the two predicted probabilities for Black testers. Half the testers are white and half are Black. Using your four predictions, compute the average marginal effect (AME) of a criminal record on the callback probability, and write the one sentence you would put in a results section to report it. [Week 4, slide 16–18]

Answer. Black, no record: $-0.66 - 1.25 = -1.91$, odds $e^{-1.91} = 0.15$, $\hat{p} = 0.13$. Black, record: $-1.91 - 0.99 - 0.29 = -3.19$, odds $e^{-3.19} = 0.04$, $\hat{p} = 0.04$. AME $= \frac{1}{2}(0.16 - 0.34) + \frac{1}{2}(0.04 - 0.13) = \frac{1}{2}(-0.18) + \frac{1}{2}(-0.09) = -0.135$. Sentence: "Averaged over the testers, a criminal record lowers the probability of a callback by about 13.5 percentage points." Full credit for any sentence that states the sign, the size in probability points, and that it is an average over the sample; a good answer adds that the drop is 18 points for white testers and 9 for Black testers.

Question 3: Military Service and Earnings (Angrist 1990, 1998; 20 points)

What did military service do to later earnings? Two papers, two designs.

The draft lottery. Angrist (1990) uses Social Security earnings records for white men born in 1950. The 1970 draft lottery gave every birthdate a random number; men whose number fell below a ceiling were eligible to be drafted. $Z_i = 1$ if eligible, $D_i = 1$ if the man served, $Y_i =$ 1981 earnings in current dollars. The first two columns give the share who served among the eligible, $\Pr(D_i = 1 \mid Z_i = 1)$, and among the not eligible, $\Pr(D_i = 1 \mid Z_i = 0)$.

	share who served		first stage	reduced form
	among eligible	among not eligible		(1981 $)
estimate	0.353	0.193	0.159	−435.8
(se)			(0.040)	(210.5)

Matching on applicants. Angrist (1998) studies the all-volunteer army: men who applied to enlist in 1979–82, with their 1988–91 earnings. Applicants who served are compared with applicants who did not, within cells of birth year, schooling, and score on the military's entrance test. The matching estimate is the effect of service on those who served (the ATT): what veterans earned compared with what the same men would have earned had they not served; the cells are weighted by their share of veterans. The regression estimate is OLS of earnings on veteran status with a dummy for every cell (a saturated regression).

	mean earnings	difference in means	matching	regression
whites	14,537	1,233 (60)	−197 (71)	−89 (63)
non-whites	11,664	2,449 (47)	840 (63)	1,074 (51)

(a) Among white applicants, veterans earn $1,233 more than non-veterans, yet the matching estimate is −$197. Explain in a few sentences why the two numbers differ, what the gap between them measures, and in which direction applicants who served were selected. [Week 5, slide 10] [Week 6, slide 11]

Answer. The difference in means compares veterans with non-veterans who differ in birth year, schooling, and test score; the matching estimate compares them within cells of those variables. The gap, $1{,}233 - (-197) = \$1{,}430$, is selection bias: applicants who served would have earned about $1,430 more than those who did not even without serving. The military screens applicants on test scores, schooling, and health, so veterans were positively selected; once that is removed, service itself slightly lowered white veterans' earnings.

(b) Matching and regression use the same cells yet give −197 and −89. Say how each weights the cells, and why the two numbers need not agree. [Week 6, slide 70, 76–77]

Answer. Matching computes the ATT: it keeps every veteran, reweights the non-veterans within each cell to look like the veterans, and weights the cell effects $\tau(x)$ by the cells' share of veterans, $\Pr(X = x \mid D = 1)$, which is proportional to $n_x\, e(x)$. Regression with cell dummies weights each cell by $n_x\, e(x)(1 - e(x))$, the variance of veteran status in the cell, so it leans toward cells where veterans and non-veterans are

mixed and away from cells that are almost all veterans. The two agree only if $\tau(x)$ is constant across cells; here they differ by \$108.

(c) From the lottery table, compute the IV (Wald) estimate and interpret it in one sentence. [Week 7, slide 9–10, 34]

Answer. $\hat{\beta}_{\text{Wald}}$ = reduced form/first stage = $-435.8/0.159 = -2{,}741$. Draft eligibility lowered 1981 earnings by \$436 across all eligible men; scaled by the 16 points of service it induced, serving cost the men who served because they were drafted about \$2,700 a year. (Week 7 slide 34 adds "about 15% of earnings," using mean 1981 earnings of \$16,461, which the exam does not print; do not require it.)

(d) State the three IV conditions for draft eligibility as an instrument for service. For each, say whether it is plausible here and why. Then give one concrete threat to one of the conditions. [Week 7, slide 6–7, 33]

Answer. Relevance: eligibility must move service. Plausible and confirmed in the table: the share who served rises from 0.193 to 0.353, a first stage of 0.159 (0.040). Independence: eligibility must be unrelated to everything else that determines earnings. Plausible because the lottery number is a random draw from birthdates, so it holds by design. Exclusion: eligibility must affect earnings only through service. Plausible but not guaranteed by the lottery; it is an argument, not a test. One threat (to exclusion): eligible men stayed in school or enrolled in college to keep a deferment, which changes earnings without any service; or employers invested less in men who might be called up.

(e) Describe compliers, always-takers, and never-takers in the lottery setting. What does monotonicity rule out, and is it plausible here? [Week 7, slide 12–13]

Answer. Compliers: served because they were eligible and would not have volunteered. Always-takers: volunteers, who served whatever their number. Never-takers: men who did not serve either way (failed the physical, obtained deferments, or avoided the draft). Monotonicity rules out defiers, men who served only because they were *not* eligible. Plausible: a low lottery number can only push a man toward service, never away from it.

Bonus: Where the Naive Comparison Goes Wrong

Ten points. Use only the definitions of the ATE, ATT, ATU, and $\pi = \Pr(D_i = 1)$, the switching equation $Y_i = D_i Y_i(1) + (1 - D_i)Y_i(0)$, and, where stated, the identity $\text{ATE} = \pi\,\text{ATT} + (1 - \pi)\,\text{ATU}$. [Week 5, slide 9–11]

(a) Show that the naive comparison is the ATU plus a selection-bias term (hint: add and subtract $E[Y_i(1) \mid D_i = 0]$):

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \text{ATU} + \underbrace{E[Y_i(1) \mid D_i = 1] - E[Y_i(1) \mid D_i = 0]}_{\text{selection bias}}.$$

Say in one sentence what the selection-bias term compares.

Answer. By the switching equation, $E[Y_i \mid D_i = 1] = E[Y_i(1) \mid D_i = 1]$ and $E[Y_i \mid D_i = 0] = E[Y_i(0) \mid D_i = 0]$. Add and subtract $E[Y_i(1) \mid D_i = 0]$:

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(1) \mid D_i = 0]\big) \\ &\quad + \big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big) \\ &= \underbrace{E[Y_i(1) \mid D_i = 1] - E[Y_i(1) \mid D_i = 0]}_{\text{selection bias}} \\ &\quad + \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 0]}_{\text{ATU}}. \end{aligned}$$

The selection-bias term compares the treated and the untreated in the *treated* state, $Y_i(1)$: would the treated have done better than the untreated even if everyone had been treated? It is the mirror image of Week 5's decomposition around the ATT, which compares the two groups in the untreated state, $Y_i(0)$.

(b) Now assume $\text{ATE} = \pi\,\text{ATT} + (1 - \pi)\,\text{ATU}$. Prove that

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \text{ATE} &+ \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline selection}} \\ &+ \underbrace{(1 - \pi)(\text{ATT} - \text{ATU})}_{\text{selection on the effect}}. \end{aligned}$$

(Hint: first add and subtract $E[Y_i(0) \mid D_i = 1]$ so that the first two terms form the ATT; then use the assumed identity to write $\text{ATT} = \text{ATE} + (1 - \pi)(\text{ATT} - \text{ATU})$.)

Answer. Add and subtract $E[Y_i(0) \mid D_i = 1]$:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline selection}}.$$

From $\text{ATE} = \pi\,\text{ATT} + (1-\pi)\,\text{ATU}$, subtract ATT from both sides: $\text{ATE} - \text{ATT} = (1-\pi)(\text{ATU} - \text{ATT})$, so $\text{ATT} = \text{ATE} + (1-\pi)(\text{ATT} - \text{ATU})$. Substituting for the ATT gives the three terms. The treated can differ from the untreated in two ways: in what they would have had anyway ($Y_i(0)$), and in how much they gain (τ_i).

(c) Treatment is now assigned by lottery. Show that the selection-bias term in (a) and both selection terms in (b) are zero, so that the naive comparison equals the ATE (which then also equals the ATT and the ATU).

Answer. Random assignment gives $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$, so conditioning on D_i changes no expectation of a potential outcome: $E[Y_i(1) \mid D_i = 1] = E[Y_i(1) \mid D_i = 0]$, which zeroes the term in (a); $E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0]$, which zeroes baseline selection; and $E[\tau_i \mid D_i = 1] = E[\tau_i \mid D_i = 0]$, so ATT = ATU and selection on the effect is zero. Then $\text{ATE} = \pi\,\text{ATT} + (1-\pi)\,\text{ATU} = \text{ATT} = \text{ATU}$, and the difference in means is unbiased for all three.

Reference: Computed Values

Use this page instead of a calculator. Write the expression you need in your answer, for example $e^{-0.66}$, then copy its value from here. Values are rounded. The list includes some values the exam does not use.

x	e^x	$e^x/(1+e^x)$
−3.19	0.041	0.040
−2.90	0.055	0.052
−2.20	0.111	0.100
−1.94	0.144	0.126
−1.91	0.148	0.129
−1.65	0.192	0.161
−1.28	0.278	0.218
−0.99	0.372	0.271
−0.66	0.517	0.341
−0.392	0.676	0.403
−0.29	0.748	0.428
0.29	1.336	0.572
0.92	2.509	0.715
0.99	2.691	0.729

expression	value
0.062×0.271	0.017
0.108×0.271	0.029
$435.8/0.159$	2,741
435.8×0.159	69.3
$0.159/435.8$	0.0004
$12/40$	0.30
$0.01/0.2$	0.05
0.01×0.2	0.002